TrialMatch

AgentVerse 2026 · Biotech track

1. The problem

Lung cancer causes around 32,800 deaths a year in the UK, and only 34% of cases in England with a known stage are diagnosed at stage I or II[1]. For advanced non-small cell lung cancer (NSCLC), a trial is often one of the best remaining options, yet **about one in five adult cancer trials fails to complete**, with **poor accrual behind 39%** of those failures[2]. Matching is hard: the median number of eligibility criteria in thoracic oncology trials rose from 16 to 27 between 1986–95 and 2006–16[3], molecular results can take four weeks after biopsy[4], and trials needing a fresh biopsy see **49.1% screen failure against 26.5%**[5]. Black, Asian and Hispanic patients remain under-represented in lung cancer trials[6].

2. Who it helps

Patients with advanced NSCLC get a short list of best-fitting trials and see why each fits; **research nurses** prescreen candidates with the evidence for every criterion; **trial sites** gain faster, more representative recruitment. **A match needs the patient's interest and the nurse's pass.**

3. Our data

Trials: Trial Scout searches ClinicalTrials.gov[7] (API v2) live for **recruiting Phase III NSCLC trials with UK sites**. The demo and the offline test use **19 synthetic trials** (LT-301 to LT-319) whose entry rules are written as code. **Patients are synthetic** (demo accounts, a SMART on FHIR sandbox patient, made-up clinic letters); missing values stay "not recorded". An earlier prototype matched 30 de-identified cBioPortal records[8] against 205 real trials.

4. How the agent works

TrialMatch shortlists recruiting Phase III trials for a person with advanced lung cancer: **entry rules decide, and a score only orders what is left**. Lung cancer comes first because it has the best-researched data; the approach extends to other diseases. The **Records Reader** pulls facts from a clinic letter or FHIR record, and the **Skeptic** drops any fact whose quote is not in the source word for word. The **Matcher** checks each rule in fixed code: met, not met, or **"to check" when a fact is missing**. For live trials, a model **Pre-check** (MedGemma, or Gemini on the hosted demo) reads the full entry rules and names the fact that would settle each unknown. **Find and check** asks the patient what settles the most trials, checks again, and widens the search if all are ruled out.

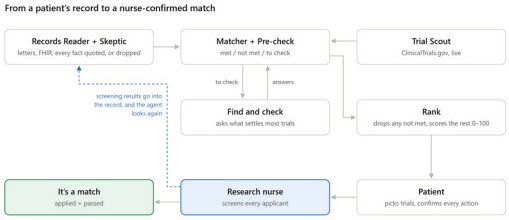


Figure 1. Agent pipeline: a question loop with the patient and a screening loop with the research nurse.

Ranking: a trial with any rule not met is removed, with its reasons. The rest score 0–100: **35% biomarker fit, 20% timing, 20% practicality** (travel, visits, randomisation), **15% earlier evidence and 10% certainty**. A trial past the patient's travel limit is hidden, not ruled out: **preferences never change eligibility**.

Screening loop: the nurse moves each applicant through prescreen, screening and enrolment. Results go into the patient's record, and if a rule fails, their agent looks for other trials. Chat answers stay marked as the patient's own until the care team confirms them.

Agents in the chat: a Gemini agent looks up live trials and offers actions (apply, pass, withdraw, apply-for-me, update the record; trial teams can invite people who could fit). Code blocks unsafe actions, such as applying to a trial the patient fails, and **nothing happens until the person taps to confirm**. A Research agent summarises Europe PMC papers, citing one in every sentence. Trials show their drug target in 3D from the Protein Data Bank, and a **local mode runs every AI step on MedGemma**.

Table 1. Worked example: demo account Mei L. (stage IV adenocarcinoma, EGFR exon 19 deletion, growing on osimertinib, PD-L1 10%, ECOG 1, brain scan not recorded, London, travels up to 25 miles).

Trial	Entry rules	Shown to her	Score
LT-311: HER3-targeted antibody–drug conjugate, London	**8 of 8 met**	Strong match	**97**
LT-301: new tablet after EGFR treatment, London	8 met, 1 to check (brain)	Possible match	97
LT-304: immunotherapy for high PD-L1	3 not met (PD-L1, EGFR, earlier treatment)	Ruled out, with reasons	—

A registry search lists all 18 NSCLC trials for her. TrialMatch rules out 6, hides 1 past her travel limit and finds **4 with every rule met**; the other 7 all wait on whether a scan has shown spread to the brain, so **her agent asks that first**.

5. How we evaluated it

- **Matching:** on 45,600 synthetic patient–trial pairs (2,400 patients × 19 trials, labels from rules we wrote)[9], a plain registry search got **93% of its "eligible" matches wrong**. TrialMatch got **none of its "all rules met" matches wrong**; counting "to check", it cut trials to review per patient **from 16.7 to 1.6** and still found every real match.
- **Tests:** automated tests of the Skeptic, the pre-check and the chat agent's tools and actions pass against a stubbed network, not the live model. An earlier prototype's parser got **92% (58/63)** of fields right on 8 held-out real trials[9].

6. Limits and safeguards

- **Human in the loop:** every fact quotes its source; a research nurse screens every applicant; the chat only offers actions, and **the tool never enrols anyone**.
- **Assumptions labelled:** the score weights are our own, not validated against real enrolment; **results are a demonstration on synthetic data, not real-world performance**. Known gaps: an unrecorded stage gives "not met" rather than "to check" (fix in progress), and a silent yes/no fact reads as "no". HER2, NTRK and MET amplification are not modelled.
- **Data limits:** demo trials and patients are synthetic and written by us; the registry search, Europe PMC and the FHIR test server are real public services.
- **Privacy and status:** demo data stays in each browser. The hosted demo's AI runs on Gemini in Google's cloud and says so, so only synthetic data is used; in local mode, MedGemma runs on the user's computer and the record never leaves it. Not a medical device.

References **1.** Cancer Research UK, lung cancer statistics. **2.** Stensland KD et al., J Natl Cancer Inst 2014;106(9). **3.** Garcia S et al., J Thorac Oncol 2017;12(10):1489–95. **4.** Mangera Z, Hospital Healthcare Europe. **5.** Spiegel ML et al., Cancer 2017;123(24):4800–7. **6.** Pang HH et al., J Clin Oncol 2016. **7.** ClinicalTrials.gov API v2, US NLM. **8.** cBioPortal datahub (TCGA LUAD, LUSC; MSKCC NSCLC 2018). **9.** Trial Match team, repository evaluation, 2026.